

MH1101 Tutorial 2 (Week 3)

Reference: Sections 1.3 1.4, 1.5 (Lecture Notes)

1. Use the Fundamental Theorem of Calculus Part 1 to find the derivative of the following functions:

(i) $g(x) = \int_1^x \cos(t^2) dt$

(ii) $F(s) = \int_s^0 \sqrt{1 + \sec t} dt$

(iii) $R(y) = \int_{2y}^{3y} \frac{u^2 - 1}{u^2 + 1} du$

2. Evaluate the following definite integral:

(i) $\int_{-1}^2 (x - 2|x|) dx$

(ii) $\int_{-2}^2 f(x) dx$, where

$$f(x) = \begin{cases} x^3 & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

3. Express the given indefinite integrals in terms of the function f :

(i) $\int f'(5x) dx$

(ii) $\int x f'(3x^2) dx$

4. Evaluate the integral by making a suitable substitution.

(i) $\int x\sqrt{1-x^2} dx$

(iv) $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$

(ii) $\int x^2 \sin(x^3) dx$

(v) $\int \frac{1}{\cos^2 t \sqrt{1+\tan t}} dt$

(iii) $\int_{-5/2}^{-2} x(2x+5)^8 dx$

(vi) $\int_{\pi/3}^{2\pi/3} \csc^2\left(\frac{t}{2}\right) dt$

5. If f is continuous on $[0, 1]$, show that

$$\int_0^\pi x f(\sin x) dx = \frac{\pi}{2} \int_0^\pi f(\sin x) dx.$$

Hint: Substitute $u = \pi - x$.

Hence, evaluate the integral:

$$\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx.$$

You may assume that $\int \frac{1}{1+x^2} dx = \tan^{-1} x + C$.

6. Let $f(x) = \frac{1}{\sqrt{x}}$.

(i) Find the average value of f on the interval $[1, 4]$.

(ii) Find the value of c guaranteed by the Mean Value Theorem for Integrals such that $f_{ave} = f(c)$.

7. If $f_{ave}[a, b]$ denotes the average value of f on the interval $[a, b]$ and

$a < c < b$, show that

$$f_{ave}[a, b] = \left(\frac{c-a}{b-a} \right) f_{ave}[a, c] + \left(\frac{b-c}{b-a} \right) f_{ave}[c, b].$$

Answer Keys. 1. (i) $\cos(x^2)$ (ii) $-\sqrt{1 + \sec s}$ (iii) $2 \left(-\frac{4y^2-1}{4y^2+1} \right) + 3 \left(\frac{9y^2-1}{9y^2+1} \right)$
 2. (i) $-7/2$ (ii) 6 3. (i) $\frac{1}{5}f(5x)+C$ (ii) $\frac{1}{6}f(3x^2)+C$ 4. (i) $-\frac{1}{3}(1-x^2)^{3/2}+C$
 (ii) $-\frac{1}{3}\cos(x^3)+C$ (iii) $-\frac{41}{360}$ (iv) $-2\cos(\sqrt{x})+C$ (v) $2\sqrt{1+\tan t}+C$ (vi)
 $\frac{4}{3}\sqrt{3}$ 5. $\frac{\pi^2}{4}$.
 6. (i) $\frac{2}{3}$ (iii) $\frac{9}{4}$